

Exponent Laws with Variable Bases

Teacher's Edition

Read This First

This is the first of the two Thread A days placed inside Unit 3 under ruling 14: MA.8.AR.1.1 (Math Nation Unit 14, lessons 14.1 and 14.2) taught right after the numerical laws are complete and before scientific notation. The content is the same seven laws with variable bases, and the lesson is built to say so out loud — every law is introduced by the numerical version the class already owns.

Math Nation's Unit 14 pages were not part of the Unit 3 package, so the items here are original, aligned to the benchmark text, the B1G-M's example $((3x^3y^{-2})^3)$ — held for tomorrow, because of the negative exponent), its listed items $(x^5x^8 = x^{13})$ and its task (Rachel and Justina, whiteboard 9). The three IXL skills for the day are the ones the Math Nation plan attaches to 14.1 and 14.2.

Every quotient today lands on a positive or zero exponent by design; negative results are tomorrow's lesson. Do not let a student 'discover' x^{-2} today — say 'tomorrow' and move on.

The word monomial is used on the student pages. It is defined in the vocabulary box and on the reference sheet; the benchmark's boundary is 'monomial bases', so the word earns its place.

Standard and Targets

MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

Benchmark Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

Boundary Integer exponents; monomial bases. Today's exponents are whole numbers; the negative ones arrive on day 2. Rational exponents are Algebra 1.

Learning target I can multiply and divide monomials, and raise a monomial to a power, by applying the laws of exponents to variable bases.

Essential question Do the laws of exponents change when the base is a variable?

Building on MA.7.NSO.1.1 and MA.8.NSO.1.3 (3.01–3.07) — the laws with numerical bases; Grade 7 — $x \cdot x \cdot x = x^3$ and $3x$ means $3 \cdot x$.

Working toward T-A2 — negative exponents on variable bases and several laws at once; MA.912.NSO.1.2 — rational exponents on algebraic expressions in Algebra 1.

Where this sits in the course First of two MA.8.AR.1.1 days. The Grade 8 FAST reports this under Algebraic Reasoning, not Number Sense — the same laws, a different reporting category. The B1G-M misconceptions for this benchmark are the numerical ones in new clothes: multiplying the exponent into the base ($5 \cdot 2$ for 5^2),

adding when the law says multiply, and forgetting that a coefficient inside parentheses is raised too.

Lesson at a Glance

Windy Hill periods run **53 minutes**. The whiteboard round takes the remainder after the taught blocks — **19 minutes** in this lesson — and IXL takes the last 5. Slide numbers below are read from the deck when this edition is built.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-6	11 min
Example 1	7-8	3 min
Your Turn	9-10	3 min
Example 2	11-12	3 min
Your Turn	13-14	3 min
Whiteboards	15-32	19 min
IXL	33	5 min
Total		53 min

Materials Whiteboards and markers. Calculators are allowed on everything; there is little for them to do today beyond 15^3 .

Where the lesson lives Notes II — 'everything inside the parentheses gets the exponent, the number included'. Whiteboard 5 tells you whether it landed; whiteboard 9 asks for it in writing.

No partner or group work anywhere in this lesson. Every Math Nation exploration this lesson draws on was written for pairs; here the same tables are worked from the front and on individual boards.

Vocabulary

monomial — a number, a variable, or a product of a number and variables with whole-number exponents: 7 , x , $-3x^2y^5$

coefficient — the number factor of a monomial: in $-3x^2y^5$ the coefficient is -3

like bases — powers of the same variable: x^2 and x^5 have like bases; x^2 and y^5 do not

Warm-Up — spaced retrieval

Four questions, two minutes silent, then the reveal. No reteach longer than one sentence per question.

1. Find the value: $(2^3)^2 \cdot 2^{-4}$ **4**

retrieves: yesterday — 3.06–3.07 — several laws in one numerical expression: $2^6 \cdot 2^{-4} = 2^2$

2. Write as one power: $\frac{6^7}{6^5}$ **$6^2 = 36$**

retrieves: last week — 3.02 — quotient of powers

3. $\sqrt{50}$ is between which two whole numbers? **7 and 8**

retrieves: last unit — Unit 2 — locating a square root between consecutive integers (2.05):
 $49 < 50 < 64$

4. Write with an exponent: $y \cdot y \cdot y \cdot y \cdot y$ **y^5**

retrieves: prior grade — the prerequisite this lesson needs — Grade 7 (Unit 1) — a repeated variable factor is a power, for the same reason $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$ is 2^5

Slide-by-Slide Teaching Notes

Slide 1. Exponent Laws with Variable Bases 1 min

- Post the learning target. Say the 'today' line and nothing else yet.

Slide 2. Warm-Up — On your own. Four minutes. No notes. 4 min

- Q4 is the bridge: $y \cdot y \cdot y \cdot y \cdot y$ is y^5 for exactly the reason $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$ is 2^5 . A student who can write that can do every law in this lesson by expanding. Q1 is yesterday's skill and the same skill as today's with the letters swapped in.

Slide 3. Warm-Up — Answers. 1 min

- Reveal. Ask for the band each question came from only if time allows.

Slide 4. Notes — Copy both lines. 4 min

- Expand once, from the front: x^2 is $x \cdot x$, x^3 is $x \cdot x \cdot x$, so the product is five x's. The law is the one they know; the only change is that nothing gets 'evaluated' at the end — x^5 is the answer. Then the coefficients: $3x^2 \cdot 5x^4$ is $3 \cdot 5 \cdot x^2 \cdot x^4$. Coefficients multiply; exponents add. Say the two wrong boards out loud before they appear: $15x^8$ (multiplied the exponents) and $8x^6$ (added the coefficients).
- **OFF THE SLIDE, YOURS TO SAY:** 'Coefficients multiply. Exponents add. Two different operations in the same problem.'

Slide 5. Notes 4 min

- $(x^3)^4$ is four copies of x^3 , twelve x's: multiply. Then the power of a product, where the coefficient is the thing students forget to raise: $(2y^2)^3$ is $2y^2 \cdot 2y^2 \cdot 2y^2$, so the 2 is cubed too. The state guide's task is the pair Rachel/Justina: $(15xy^2)^3$ expanded as $15 \cdot 15 \cdot 15 \cdot x \cdot x \cdot x \cdot y \cdot y \cdot y \cdot y \cdot y$ and written as $3,375x^3y^6$ are the same expression. Both are correct; one is finished.
- **OFF THE SLIDE, YOURS TO SAY:** 'Everything inside the parentheses gets the exponent — the number included.'

Slide 6. Notes *3 min*

- x^7 over x^2 cancels two x's and leaves five: subtract. Coefficients divide as numbers; each variable is handled on its own. Today every quotient comes out with positive exponents (or a zero exponent, which is 1) — tomorrow the subtraction is allowed to go below zero. Note the last line: n^4/n^4 is $n^0 = 1$, not 0 and not n .

Slide 7. Example 1 — Multiplying two monomials. *1 min*

- SPOKEN ONLY. Three kinds of factor in each monomial: a coefficient, x's and y's. Ask what happens to each kind before anyone writes.

Slide 8. Example 1 — Worked. *2 min*

- Regroup by kind, then apply the product law twice. The y in the first factor has exponent 1 — a board with y^5 forgot it. A board with $-8x^6y^5$ multiplied the exponents; a board with $2x^5y^6$ added the coefficients.

Slide 9. Your Turn — Same steps, your numbers. Boards up when done. *2 min*

- Catches the missing exponent 1 on a and the exponent-multiplier ($15a^2b^8$). $15a^3b^6$ is right.

Slide 10. Your Turn — Answer. *1 min*

- Reveal; name what a wrong board most likely did (see the note above).

Slide 11. Example 2 — A power of a product, then a quotient. *1 min*

- SPOKEN ONLY. Two laws in order: the numerator is a power of a product, then divide. Ask which law comes first and why (the exponent outside the parentheses has to be dealt with before anything can be divided).

Slide 12. Example 2 — Worked. *2 min*

- Raise every factor in the numerator: $3^3 = 27$, $(m^2)^3 = m^6$, n^3 . Then divide the coefficients and subtract each variable's exponents. A board with $3m^6n^3$ on top never cubed the 3; a board with $3m^2n^3$ forgot the n in the denominator.

Slide 13. Your Turn — Same steps, your numbers. Boards up when done. *2 min*

- Catches the un-raised coefficient ($2p^{12}/8p^5 = p^7/4$) and the exponent-adder ($2p^9$). $2p^7$ is right.

Slide 14. Your Turn — Answer. *1 min*

- Reveal; name what a wrong board most likely did (see the note above).

Slide 15. Whiteboards · Question 1 of 9 *19 min for the round*

- The product law, plain, with a letter. The state guide's own item. A board with x^{40} multiplied the exponents.

Slide 16. Whiteboards · Question 1 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 17. Whiteboards · Question 2 of 9

- A board with y^7 added. Ask that student to write out three copies of y^4 .

Slide 18. Whiteboards · Question 2 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 19. Whiteboards · Question 3 of 9

- The coefficient is the thing that gets forgotten. $2x^{12}$ never raised the 2; $8x^{12}$ multiplied 2 by 4 — the guide's 'multiplying the exponent into the base' ($6^3 = 18$) in algebraic clothing.

Slide 20. Whiteboards · Question 3 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 21. Whiteboards · Question 4 of 9

- Two variables and an invisible exponent 1 on a and on b. $7a^3b^6$ added the coefficients; $12a^2b^5$ ignored the exponent-1 factors.

Slide 22. Whiteboards · Question 4 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 23. Whiteboards · Question 5 of 9

- Diagnostic. B and D are the two coefficient errors; C is the exponent error. Three different repairs, so read the letters, not the count.

Slide 24. Whiteboards · Question 5 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 25. Whiteboards · Question 6 of 9

- Diagnostic for the quotient. B is the one to expect: 'divide' applied to both the coefficients and the exponents.

Slide 26. Whiteboards · Question 6 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 27. Whiteboards · Question 7 of 9

- The zero exponent hiding in a quotient. A board with m^4n read n^0 as n ; a board with 0 read it as 0 — the 3.02 misconception with a letter.

Slide 28. Whiteboards · Question 7 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 29. Whiteboards · Question 8 of 9

- Working backwards — the shape the guide expects and the book does not ask. A board with 33 or with $11/3$ tried to multiply or divide.

Slide 30. Whiteboards · Question 8 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 31. Whiteboards • Question 9 of 9

- Written. The state guide's task for this benchmark. Full credit: 'both', with the reason that Rachel expanded the power and Justina simplified the same product. A student who picks Justina alone has not seen that the two are the same expression; one who picks Rachel alone thinks the 15 should not be cubed.

Slide 32. Whiteboards • Question 9 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 33. IXL — Last five minutes. 5 min

- IXL: whatever is not finished is tonight's practice; SmartScore 67.

The Whiteboard Round — what each board tells you

Nine questions, one at a time, each with its reveal. Boards down until the cue, all up together, scan the back row first, question the blank boards before the wrong ones. Question 9 is written work. Every wrong option on a multiple-choice question is a named error; the benchmark it comes from is cited.

#	Question	Answer	What a wrong board says
1	$x^5 \cdot x^8$	x^{13}	x^{40} — multiplied the exponents [B1G-M 8.AR.1.1 misconception: misapplied laws]; $2x^{13}$ — an invented coefficient
2	$(y^4)^3$	y^{12}	y^7 — added the exponents [8.AR.1.1: power of a power confused with a product]
3	$(2x^3)^4$	$16x^{12}$	$2x^{12}$ — the coefficient was not raised [B1G-M 8.AR.1.1: power of a product]; $8x^{12}$ — coefficient multiplied by the exponent [B1G-M misconception: multiplying the exponent into the base]
4	$(3a^2b)(4ab^5)$	$12a^3b^6$	$7a^3b^6$ — coefficients added instead of multiplied [8.AR.1.1: coefficients are factors]; $12a^2b^5$ — the factors a and b with exponent 1 were dropped
5	Which expression is equivalent?	$A - 25m^6$	(B) $5 \cdot 2$ instead of 5^2 — the exponent was multiplied into the coefficient [B1G-M 8.AR.1.1 misconception: multiplying the exponent into the base]; (C) $3 + 2$ instead of $3 \cdot 2$ [power of a power confused with a product of powers]; (D) the coefficient was not raised at all [8.AR.1.1: power of a product]

#	Question	Answer	What a wrong board says
6	Which expression is equivalent?	$A - 4x^6$	(B) $8 \div 2$ — the exponents were divided instead of subtracted [B1G-M 8.AR.1.1: misapplied quotient law]; (C) $12 - 3$ — the coefficients were subtracted instead of divided [8.AR.1.1: coefficients are factors]; (D) $8 + 2$ — added the exponents [quotient law confused with product law]
7	$\frac{m^6 n^4}{m^2 n^4}$	m^4	$m^4 n$ — n^0 read as n [8.AR.1.1: zero exponent]; 0 or $m^4 \cdot 0$ — n^0 read as 0 [B1G-M 8.NSO.1.3 misconception: $a^0 = 0$]
8	Find the value of n : $x^3 \cdot x^n = x^{11}$	$n = 8$	14 — added 3 to 11; 33 — multiplied [8.AR.1.1: product law is additive in the exponents]
9	Rachel writes $(15xy^2)^3$ as $15 \cdot 15 \cdot 15 \cdot x \cdot x \cdot x \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y$. Justina writes it as $3,375x^3y^6$. Who is correct? Explain.	Both are correct; $15^3 = 3,375$, $x \cdot x \cdot x = x^3$, and six y's make y^6.	'Justina only' — the expanded form is not recognized as equivalent [8.AR.1.1: generate equivalent expressions]; 'Rachel only' or ' $45x^3y^6$ ' — the coefficient was not cubed [B1G-M 8.AR.1.1: power of a product]

Question Bank — variation and answers

Variation. Questions 1–6 are one law each: product, power of a power, power of a product, quotient, a negative coefficient, two variables in a quotient. 7–8 combine two laws in the two orders that occur (multiply then raise; raise then divide), and vary whether a zero exponent appears. 9 works backwards to a missing exponent in each of the three laws; 10 is the geometric setting, where the square is the trap (the coefficient is squared). 11 is the diagnostic multiple choice; 12 asks for equivalence judged four ways with two non-equivalent options wrong for two different reasons. The Additional sheet mirrors each position.

The bank and its Additional sheet are posted, never assigned; they feed the quizzes. Answers below are the same values the keys print — both come from one source.

Question Bank

- x^{13}
- n^{30}
- $27y^{12}$
- a^6
- $-12x^9$
- $4p^3q^2$
- (a) $10x^5y^5$ (b) $16m^4n^6$ (c) a^7b^7

8. (a) $3m^2n^2$ (b) y^3 (c) $2k^5$
9. (a) $n = 8$ (b) $n = 5$ (c) $n = 7$
10. (a) $12x^7$ (b) $16y^{10}$
11. A A
12. A, B, C, D A, B, C and D

Question Bank — Additional

1. y^{13}
2. k^{24}
3. $125x^6$
4. b^7
5. $12a^7$
6. $3xy^3$
7. (a) $18a^5b^5$ (b) $8p^9q^{15}$ (c) x^6y^7
8. (a) $2x^5y^3$ (b) m^3 (c) $5c$
9. (a) $n = 7$ (b) $n = 6$ (c) $n = 10$
10. (a) $27a^7$ (b) $49b^6$
11. A A
12. A, B, C, D A, B, C and D

What the audit against Math Nation found

- All 55 items on the bank, additional bank, warm-up, examples and round were re-derived by sympy at build time with the variables declared as symbols (mathcheck: 0 findings).
- The B1G-M items quoted: $x^5x^8 = x^{13}$ (whiteboard 1, verified); the Rachel/Justina task $(15xy^2)^3 = 3,375x^3y^6$ (whiteboard 9, verified $15^3 = 3,375$).
- The B1G-M example $(3x^3y^{-2})^3 = 27x^9y^{-6}$ contains a negative exponent and is held for day 2.
- No Math Nation Unit 14 pages were available for audit; nothing here is copied from them.

What changed from the book

- Math Nation 14.1 and 14.2 (two book lessons) → one period under ruling 14, since the numerical laws were completed yesterday and each law needs only its variable-base restatement.
- Book-style partner explorations → Notes I-III from the front, with expansion shown once per law.
- Cool-downs → whiteboard round; the state guide's Rachel/Justina task → whiteboard 9, written.